# Akoizo Original Practice - 2027 Rotation Form 2 - Answer key

### Division B | Water Quality | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Explain why this transect is an estuary rather than a freshwater-only habitat.**

Synthetic estuary transect: river end salinity 0, marine end salinity 24 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.4 to 4.4 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.6 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: It connects river freshwater with seawater and has a salinity gradient produced by mixing. (2 points).

**2. (2 points) Explain how a nutrient pulse can lead to lower overnight oxygen.**

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answer: Nutrients can increase production; respiration and decomposition consume oxygen. Nighttime photosynthesis is absent, and mixing also affects the budget. (2 points).

**3. (2 points) Identify one watershed action that could reduce nutrient loading and explain its pathway.**

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answer: For example, reduce excess fertilizer and retain vegetated buffers to limit nutrient transport in runoff; identify a source and transport pathway. (2 points).

**4. (2 points) Explain how traditional ecological knowledge could improve this monitoring plan without replacing measured evidence.**

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answer: Collaborate with knowledge holders to identify historical baselines, seasonal changes and appropriate sampling locations; evaluate observations alongside recorded measurements and respect consent/context. (2 points).

**5. (2 points) Why should a reef food-web analysis distinguish herbivores from predators?**

answer: They affect different trophic pathways; grazing regulates algae while predation affects prey populations, with indirect effects across the web. (2 points).

**6. (2 points) Distinguish a harmful algal bloom from an ordinary observation of green water.**

answer: A bloom can cause toxins or oxygen stress, but color alone does not establish harmful species, toxins or ecological damage; additional evidence is needed. (2 points).

**7. (2 points) A dark urchin has conspicuously long slender spines and grazes algae. Identify its listed group.**

answer: Long-spined black sea urchin (2 points).

**8. (2 points) A shrimp has red-and-white bands and conspicuous elongated appendages. Identify the listed common-name group.**

answer: Banded coral shrimp (2 points).

**9. (2 points) An elongate, soft-bodied echinoderm processes sediment on the seafloor. Identify its listed group.**

answer: Sea cucumber (2 points).

**10. (2 points) Name the listed Indo-Pacific bivalve group with a large shell, mantle and photosynthetic symbionts.**

answer: Giant clam (2 points).

**11. (2 points) An elongate predatory fish shelters in reef crevices and has strong jaws. Identify the listed eel group.**

answer: Moray eel (2 points).

**12. (2 points) Name the listed overgrowing non-coral macroalgal category that can compete for reef space.**

answer: Fleshy algae (2 points).

**13. (2 points) Calculate the overnight dissolved-oxygen decline.**

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answer: 3 mg/L (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , mg/L.

**14. (2 points) Calculate the BOD-bottle oxygen demand under the stated assumptions.**

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answer: 3.4 mg/L (2 points).

Numerical target 3.4; absolute tolerance 0.0068. Accepted units: , mg/L.

**15. (2 points) A mixture contains equal volumes of freshwater and seawater with salinity 24. Assuming additive volumes and conservative mixing, estimate salinity.**

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answer: 12 (2 points).

Numerical target 12; absolute tolerance 0.024. Accepted units: , .

**16. (2 points) Aragonite saturation decreases while other factors are held fixed. Explain a relevant concern for stony corals.**

answer: Lower aragonite saturation can make calcium-carbonate skeleton formation less favorable and increase dissolution risk; responses depend on organism and conditions. (2 points).

**17. (2 points) Explain why high turbidity and high dissolved-solids concentration are not interchangeable observations.**

answer: Turbidity reflects light scattering by suspended material. Dissolved substances can change chemistry without producing the same scattering signal. (2 points).

**18. (2 points) A field pH value changes between daytime and predawn. Name a biological mechanism consistent with this and one check on the inference.**

answer: Photosynthesis/respiration change dissolved CO2 and can shift pH. Check timing, calibration, temperature, depth, mixing and repeated observations before assigning a cause. (2 points).

**19. (2 points) A salinometer calibration gives reading 10 at 2% mass/volume salt and reading 30 at 6%. A sample reads 17. Use linear interpolation to estimate concentration.**

answer: 3.4 % (2 points).

Numerical target 3.4; absolute tolerance 0.0068. Accepted units: , %.

**20. (2 points) How many grams of salt are required for 600 mL of final solution at 3% mass/volume?**

answer: 18 g (2 points).

Numerical target 18; absolute tolerance 0.036. Accepted units: , g.